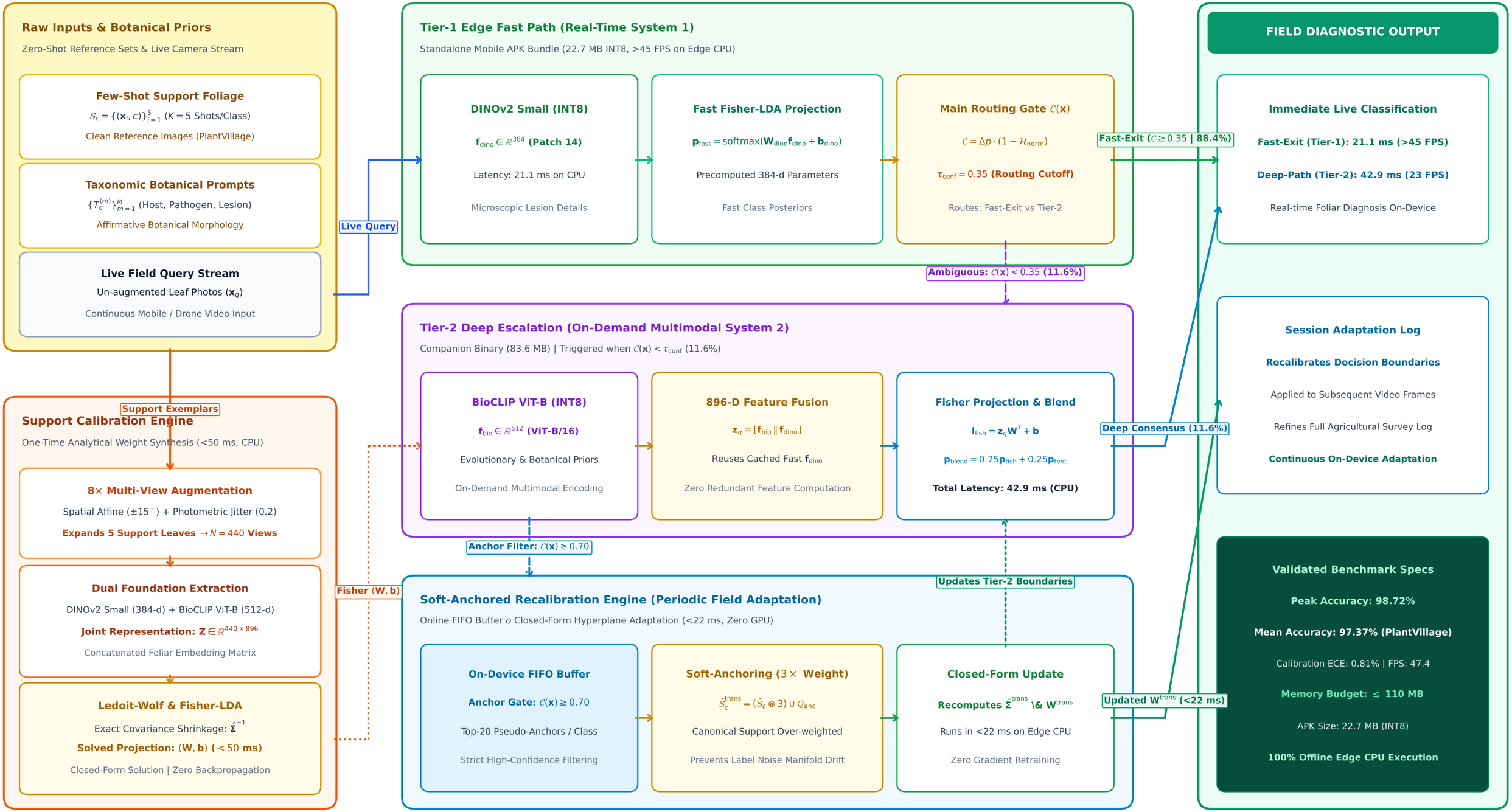

Raw Inputs & Botanical Priors
Zero-Shot Reference Sets & Live Camera Stream
Few-Shot Support Foliage
$\mathcal{S}_c = \{(\mathbf{x}_i, c)\}_{i=1}^{5}$ ($K = 5$ Shots/Class)
Clean Reference Images (PlantVillage)
Taxonomic Botanical Prompts
$\{T_c^{(m)}\}_{m=1}^{M}$ (Host, Pathogen, Lesion)
Affirmative Botanical Morphology
Live Field Query Stream
Un-augmented Leaf Photos ($\mathbf{x}_q$)
Continuous Mobile / Drone Video Input
Live Query
Support Exemplars
Support Calibration Engine
One-Time Analytical Weight Synthesis (<50 ms, CPU)
8× Multi-View Augmentation
Spatial Affine (±15°) + Photometric Jitter (0.2)
Expands 5 Support Leaves → $N = 440$ Views
Dual Foundation Extraction
DINOv2 Small (384-d) + BioCLIP ViT-B (512-d)
Joint Representation: $\mathbf{Z} \in \mathbb{R}^{440 \times 896}$
Concatenated Foliar Embedding Matrix
Ledoit-Wolf & Fisher-LDA
Exact Covariance Shrinkage: $\hat{\mathbf{\Sigma}}^{-1}$
Solved Projection: $(\mathbf{W}, \mathbf{b})$ (< 50 ms)
Closed-Form Solution | Zero Backpropagation
Fisher $(\mathbf{W}, \mathbf{b})$
Tier-1 Edge Fast Path (Real-Time System 1)
Standalone Mobile APK Bundle (22.7 MB INT8, >45 FPS on Edge CPU)
DINOv2 Small (INT8)
$\mathbf{f}_{\text{dino}} \in \mathbb{R}^{384}$ (Patch 14)
Latency: 21.1 ms on CPU
Microscopic Lesion Details
Fast Fisher-LDA Projection
$\mathbf{p}_{\text{fast}} = \text{softmax}(\mathbf{W}_{\text{dino}}\mathbf{f}_{\text{dino}} + \mathbf{b}_{\text{dino}})$
Precomputed 384-d Parameters
Fast Class Posteriors
Main Routing Gate $\mathcal{C}(\mathbf{x})$
$\mathcal{C} = \Delta p \cdot (1 - \mathcal{H}_{\text{norm}})$
$\tau_{\text{conf}} = 0.35$ (Routing Cutoff)
Routes: Fast-Exit vs Tier-2
Fast-Exit ($\mathcal{C} \geq 0.35$ | 88.4%)
Ambiguous: $\mathcal{C}(\mathbf{x}) < 0.35$ (11.6%)
Tier-2 Deep Escalation (On-Demand Multimodal System 2)
Companion Binary (83.6 MB) | Triggered when $\mathcal{C}(\mathbf{x}) < \tau_{\text{conf}}$ (11.6%)
BioCLIP ViT-B (INT8)
$\mathbf{f}_{\text{bio}} \in \mathbb{R}^{512}$ (ViT-B/16)
Evolutionary & Botanical Priors
On-Demand Multimodal Encoding
896-D Feature Fusion
$\mathbf{z}_q = [\mathbf{f}_{\text{bio}} \| \mathbf{f}_{\text{dino}}]$
Reuses Cached Fast $\mathbf{f}_{\text{dino}}$
Zero Redundant Feature Computation
Fisher Projection & Blend
$\mathbf{l}_{\text{fish}} = \mathbf{z}_q\mathbf{W}^T + \mathbf{b}$
$\mathbf{p}_{\text{blend}} = 0.75\mathbf{p}_{\text{fish}} + 0.25\mathbf{p}_{\text{text}}$
Total Latency: 42.9 ms (CPU)
Deep Consensus (11.6%)
Anchor Filter: $\mathcal{C}(\mathbf{x}) \geq 0.70$
Updates Tier-2 Boundaries
Soft-Anchored Recalibration Engine (Periodic Field Adaptation)
Online FIFO Buffer o Closed-Form Hyperplane Adaptation (<22 ms, Zero GPU)
On-Device FIFO Buffer
Anchor Gate: $\mathcal{C}(\mathbf{x}) \geq 0.70$
Top-20 Pseudo-Anchors / Class
Strict High-Confidence Filtering
Soft-Anchoring (3× Weight)
$\tilde{S}_c^{\text{trans}} = (\tilde{S}_c \otimes 3) \cup \mathcal{Q}_{\text{anc}}$
Canonical Support Over-weighted
Prevents Label Noise Manifold Drift
Closed-Form Update
Recomputes $\hat{\mathbf{\Sigma}}^{\text{trans}}$ \& $\mathbf{W}^{\text{trans}}$
Runs in <22 ms on Edge CPU
Zero Gradient Retraining
Updated $\mathbf{W}^{\text{trans}}$ (<22 ms)
FIELD DIAGNOSTIC OUTPUT
Immediate Live Classification
Fast-Exit (Tier-1): 21.1 ms (>45 FPS)
Deep-Path (Tier-2): 42.9 ms (23 FPS)
Real-time Foliar Diagnosis On-Device
Session Adaptation Log
Recalibrates Decision Boundaries
Applied to Subsequent Video Frames
Refines Full Agricultural Survey Log
Continuous On-Device Adaptation
Validated Benchmark Specs
Peak Accuracy: 98.72%
Mean Accuracy: 97.37% (PlantVillage)
Calibration ECE: 0.81% | FPS: 47.4
Memory Budget: ≤ 110 MB
APK Size: 22.7 MB (INT8)
100% Offline Edge CPU Execution